

Backbone-Modular Neuro-Symbolic Post-Correction for OCT Denoising via Explicit Predicate Maps and Stable Fuzzy Allocation

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ABSTRACT Denoisers for optical coherence tomography are trained to reduce a pixel error, so they can raise fidelity while lowering the properties a clinician reads, namely layer contrast, boundary definition and the visibility of weak structures. This paper treats the repair of those properties as a separate problem. A frozen denoiser is kept unchanged and a small policy is attached to it. The policy measures six named clinical properties in closed form, converts them into a per pixel decision through five explicit rules, and limits how far the image may move. We prove that the decision layer responds to a change in evidence by at most an exact constant below 0.75, and we measure the realised response against that constant on real images. We give three conditions under which the correction cannot lower contrast to noise ratio, and report how often they hold. The policy adds 168 630 shared trainable parameters plus a small backbone specific head to a backbone of about seven million. On the 173 held out test images, with the same rules and the same fixed weights for every backbone, the correction improves 21 of the 24 backbone and measure pairs by more than their spread across three seeds, leaves two within that spread, and lowers one, boundary sharpness on DnCNN by 1.4 percent from a starting value already above the clean reference, at a fidelity cost of 0.17 to 0.98 dB. No clinical measure falls on three of the four backbones. On two other scanners with no adaptation of any kind NAFNet and SwinIR improve every measure, while DnCNN and KNet each lose the same two measures on both scanners, 40 of 48 pairs in all.

INDEX TERMS Optical coherence tomography, image denoising, neuro-symbolic reasoning, fuzzy logic, interpretable image restoration, modular post-correction, clinical quality assessment.

I. Introduction

OPTICAL coherence tomography (OCT) is the standard non invasive way to image the retina [1]. Every OCT image carries speckle, a multiplicative disturbance that arises from coherent light scattering inside tissue [2]. Speckle blurs the border between retinal layers, lowers the contrast between tissue and background, and makes automatic layer segmentation less reliable [3].

Deep denoisers remove much of that disturbance, but they are almost always trained to minimise the squared error against a clean reference, so a denoiser can raise the peak signal to noise ratio (PSNR) of a scan while making it less useful to a clinician. Layer boundaries lose definition, the contrast between tissue and the dark vitreous falls, and thin structures with weak signal fade into the background. None of these losses is visible in

a PSNR number, and none is corrected by training the same objective harder.

A. Motivation

The gap we address is not a missing denoiser. It is that nothing sits between the denoiser and the clinical reader to state in explicit terms what a good OCT image should look like and to repair the specific ways in which a denoised image falls short. Building that part as another neural block would hide the reason for every change inside learned weights again, while a clinician, a regulator or a reviewer who sees a modified scan needs to know which property was judged deficient, where that judgement was made, and how large a change was allowed. This paper therefore treats post correction as its own problem. We keep the denoiser frozen and attach a

small policy, written as named image properties, explicit logical rules over them, and explicit limits on the size of any change, that decides where the image should change and by how much. Every learned map in the decision path is bounded, and its effect reaches the image only through a named rule.

B. New ideas in this paper

The new ideas are three, and we do not claim that edge filters, gating maps or fuzzy operators are new on their own.

- 1) Six clinical properties given in closed form. Section A defines six predicates that need no ground truth. Each returns a score for the whole image and a map of where the property is unsatisfied. Both are formulas, so the input to the rule layer can be recomputed by anyone from the released code.
- 2) A fuzzy allocation rule with a proved and measured sensitivity bound. Section B builds the allocation from Lukasiewicz connectives. Section VII defines what stability means here and proves a Lipschitz bound, and Section I measures the true sensitivity on real images and compares it with the bound. A bound that is never checked against data is a claim, not a guarantee.
- 3) A safety layer that decides how much of a proposal to keep. Section B states three constraints at three scales and shows how the number of satisfied constraints sets the fraction of the proposed change that is applied. A margin condition under which the correction cannot lower the contrast to noise ratio is proved and then tested numerically.

II. Related Work

A. Denoising of OCT images

Learned denoisers for OCT fall into three families. Convolutional networks such as DnCNN [4] and DRUNET [3] learn a residual between the noisy and the clean image. Attention based designs such as NAFNet [5] and KBNNet [9] reweight features before combining them. Transformer designs such as SwinIR [6] use windowed self attention. Classical estimators remain in use, including non local means [10] and wavelet shrinkage [11], and self supervised training removes the need for clean targets [7]. Methods built for OCT address the scarcity of paired data directly, through triplet cross fusion [12] and through sparse representation [13], [14].

All of these methods are trained to reduce a pixel error, so the quantity they optimise is not the quantity a clinician reads. Work on automated quality assessment for OCT [15], [20], [21] measures clinical quality but does not act on it. The present paper closes that loop by turning measured quality into a correction applied under stated limits.

B. Confidence and uncertainty in medical imaging

Dropout as an approximate Bayesian posterior [16], task weighting driven by learned variance [17] and deep ensembles [18] are the standard ways to attach a confidence to a prediction, and in OCT such values flag unreliable layer segmentation [3] and poor scan quality [15]. In almost all of that work the value is reported and then left alone, marking a region as doubtful without changing what happens there. We instead feed such a map into the correction decision, and Section H reports that the map we built is not a confidence in this sense.

C. Symbolic reasoning and expert networks joined with neural networks

Neuro symbolic computing [19] couples a learned model with explicit rules, and differentiable fuzzy logic makes the coupling trainable because the connectives become smooth functions of continuous truth values [8]. In medical imaging the pattern has generated spinal reports by pairing a generative network with a symbolic graph of domain knowledge [27], predicted patient deterioration with a probabilistic logic program over convolutional features [28], and interpreted corneal topography through a knowledge graph attached to a learned feature model [34]. Fuzzy membership functions have been embedded inside networks that process OCT directly, where a rough fuzzy discretisation guides layer segmentation [29], and pixel level fuzzy rules have replaced a learned attention block so that the resulting map reads as a logical statement [30].

A parallel line builds an image model from cooperating experts. Steered mixtures of experts combine local regressors through an explicit steering term [31], also for OCT speckle removal [32], and recent restoration networks route an input through experts of different capacity according to its difficulty [33]. Outside imaging, evolving fuzzy models identify actuator dynamics from data [24], learned models are combined with a physical forward operator in ultrasound computed tomography [25], and the sensitivity of a differential model to its initial conditions is treated as a source of uncertainty in simulation [26]. All of these rest on the principle we rely on, that a learned component becomes easier to trust when part of the decision is written in a language a person can read. What is still missing is a rule layer that acts on the output of a strong denoiser, names the clinical properties it protects, and limits how far it may move the image. That is the object studied here.

III. Notation and Problem Statement

A. Notation

Images are real valued arrays of height H and width W with values in $[0, 1]$. We write $\mathbf{y}$ for the noisy B scan that comes off the scanner and $\mathbf{x}$ for the clean reference obtained by averaging many repeated scans of the same

TABLE 1. Symbols used in the paper, in order of first use.

Symbol	Meaning
$\mathbf{y}$	Noisy input B scan
$\mathbf{x}$	Clean reference B scan
f_θ	Frozen denoising backbone with parameters θ
$\mathbf{b}$	Backbone output, $\mathbf{b} = f_\theta(\mathbf{y})$
ϕ	Trainable parameters of the wrapper
$\mathbf{c}$	Cooperation map with values in $[0, 1]$
$\mathbf{u}$	Demand map, $\mathbf{u} = \mathbf{1} - \mathbf{c}$
s_i	Score of clinical property i , a number in $[0, 1]$
$\mathbf{m}_i$	Failure map of clinical property i , values in $[0, 1]$
τ_i	Decision threshold for property i
Δ	Raw multiplicative gain proposed by the gain network
λ	Per pixel strength map with values in $[0, 1]$
$\mathbf{a}$	Allocation map produced by the rule layer
α	Gain after allocation, strength and clamping
$\mathbf{g}$	Background gate that suppresses change in the background
$\mathbf{v}$	Gated multiplicative change added to the backbone output
δ_e	Additive edge correction
$\mathbf{q}$	Candidate image before the safety decision
$\hat{\mathbf{x}}$	Final output of the wrapper
$\mathcal{T}, \mathcal{B}$	Sets of tissue pixels and background pixels
$\mu_{\mathcal{T}}, \mu_{\mathcal{B}}$	Mean value over tissue and over background
$\sigma_{\mathcal{B}}$	Standard deviation over background

location. A pixel index is written p , so y_p is one pixel of $\mathbf{y}$.

The symbol $\odot$ is the Hadamard product, $(\mathbf{t} \odot \mathbf{t}')_p = t_p t'_p$, and ordinary multiplication of two numbers is written by juxtaposition. Table 1 lists the symbols that recur across sections, in the order in which the paper introduces them, and a symbol used in one place only is defined in the sentence that introduces it.

B. The optimisation problem

The denoiser is given and is not changed. The task is to produce an image that beats $\mathbf{b} = f_\theta(\mathbf{y})$ on named clinical measures, stays close to it everywhere, and can explain every change it makes. The only free quantities are the wrapper parameters ϕ , namely the cooperation head, the gain network, the edge network, the strength predictor and the rule layer, while the backbone parameters θ are held fixed throughout. Let $\mathcal{D}$ be a set of paired scans

and $\hat{\mathbf{x}}(\phi)$ the output of the wrapper. We minimise the expected value of a weighted sum of five terms with fixed weights w_c , w_p , w_o and w_s ,

$$\min_{\phi} \mathbb{E}_{(\mathbf{y}, \mathbf{x}) \sim \mathcal{D}} \left[\mathcal{L}_{\text{fid}} + w_c \mathcal{L}_{\text{clin}} + w_p \mathcal{L}_{\text{pred}} + w_o \mathcal{L}_{\text{coop}} + w_s \mathcal{L}_{\text{edge}} \right], \quad (1)$$

where $\mathcal{L}_{\text{fid}}$ keeps the output near the clean reference, $\mathcal{L}_{\text{clin}}$ rewards improvement of the clinical measures over the backbone, $\mathcal{L}_{\text{pred}}$ calibrates the property scores against their thresholds, $\mathcal{L}_{\text{coop}}$ ties the amount of correction to the estimated correction demand, and $\mathcal{L}_{\text{edge}}$ supervises the additive edge branch. Section VIII gives each term in full. Three constraints must hold on every image, stated formally in Section B. The aggregate clinical quality must not fall by more than a fixed tolerance, no single property may fall by more than a fixed tolerance while another rises, and the relative change from the backbone output must stay below a fixed bound. The objective is differentiable in ϕ , including the rule layer, because every logical connective is replaced by a smooth piecewise linear surrogate, so it is solved by mini batch stochastic gradient descent with the AdamW optimiser [23]. During training the constraints enter the objective as penalties. At test time they are evaluated exactly on each image and the number satisfied decides how much of the proposed change is applied, which is what makes them binding rather than advisory.

IV. Frozen Backbone and Bounded Proposals

A. Overview

Fig. 1 shows the whole method. It runs in six stages, each answering one question, namely what the rule layer has available at each pixel, what change is available, what is wrong and where, where the change should be applied, how it is applied, and whether it is allowed at all. This section and the two that follow take the six stages in order, two to a section.

B. Stage one, the backbone and the cooperation map

The backbone is frozen after training and gives $\mathbf{b} = f_\theta(\mathbf{y})$. A two layer network h_ϕ reads a feature tensor $\mathbf{z}$ from inside it and produces the cooperation map

$$\mathbf{c} = \sigma \left(\frac{h_\phi(\mathbf{z}) + \beta_0}{\tau_0} \right), \quad (2)$$

where σ is the logistic function, $\tau_0 > 0$ is a learnable temperature and β_0 a learnable offset. The complementary demand map is $\mathbf{u} = \mathbf{1} - \mathbf{c}$. A small temperature gives a nearly binary map and a large one spreads the decision out.

Only this head depends on the backbone, because architectures expose feature tensors with different channel counts. The supplementary file lists the tensor used for each backbone. Everything after this stage is shared,

which is the precise sense in which the method is modular rather than architecture free. Section K reports what refitting the wrapper costs.

C. Stage two, the two correctors

Two changes of deliberately different kinds are proposed.

The gain network proposes a multiplicative change. It runs at quarter resolution and emits a brightness channel passed through a rectifier, so it can only brighten, and a contrast channel passed through a hyperbolic tangent, so it can go either way. Their sum is scaled by $\alpha_{\max} = 0.15$ to give the raw gain Δ . A multiplicative form suits speckle, which is itself multiplicative [2], so bright tissue receives a larger absolute change than dark background for the same relative adjustment.

The edge network proposes an additive change δ_e from four channels, namely the intensity normalised residual, the edge magnitude of the backbone output, the backbone output, and an unsharp mask. Its output is bounded to ± 0.08 and scaled by a learned factor. An additive form suits boundaries, which should be sharpened by the same amount in a bright layer as in a dark one.

A third small network predicts a per pixel strength map λ with values in $[0, 1]$ from the failure maps of Stage three. It modulates how much of the raw gain survives at each pixel.

V. Clinical Properties and Fuzzy Allocation

A. Stage three, the six clinical properties

The six properties are the interface between clinical language and the arithmetic of the method. Each names a way a denoised OCT image is known to fail and each is written as a formula so it can be recomputed by anyone. Table 2 gives, for every property, the reason it exists, where its map $\mathbf{m}_i \in [0, 1]^{H \times W}$ is large, and the threshold τ_i that separates satisfied from unsatisfied. Each score $s_i \in [0, 1]$ is a weighted sum of named subterms, given in closed form in the supplementary file. A larger score means the property is better satisfied and a larger map value means stronger local evidence that it is not. None of these quantities needs a clean reference. They are computed from the noisy input and the backbone output only, so they remain available at test time on a scanner the method has never seen, and the supplementary file shows the six maps for one image.

B. Stage four, the rule layer

The rule layer decides where the gain proposal should act through five named rules, each casting a signed vote.

Each property score is first turned into a soft failure signal by comparing it with its threshold,

$$f_i = \sigma(\kappa(\tau_i - s_i)), \quad \kappa = 10, \quad (3)$$

so f_i is near one when property i is failing and near zero when it is satisfied. The same soft comparison is used to

test the cooperation map and the strength of a proposal against their own thresholds. Each corrector produces that strength as a per pixel map from the backbone output and the failure maps it serves, large where its property is far below threshold, so it says where that corrector believes it can help. Before the rules fire, that strength is multiplied by one half plus the demand map, so the same proposal looks stronger to the rules at a pixel where the head reports low cooperation. Conjunction uses the Lukasiewicz form $\text{AND}(a, b) = \max(0, a + b - 1)$, which is piecewise linear and has a bounded slope, which is what makes the bound in Section VII tight.

The five rules are the following. Rule one fires when the cooperation map is high and the proposal is weak, and it votes against changing the image. Rule two fires when the map is low and the proposal is strong, and it votes for changing it. Rule three fires when a relevant property is failing, and it votes for changing it. Rule four fires when the map and the proposal strength are both middling, and it votes weakly for changing it. Rule five fires when several correctors want the same pixel, and it votes against changing it. Write r_1 to r_5 for the resulting truth values.

The allocation at each pixel is the bounded weighted vote

$$a = \underbrace{B_{\min} + (B_{\max} - B_{\min}) \sigma(\beta)}_{\text{base, no rule firing}} - C_1 w_1 r_1 + C_2 w_2 r_2 + C_3 w_3 r_3 + C_4 w_4 r_4 - C_5 w_5 r_5, \quad (4)$$

where β is a learnable base logit, $w_k = \sigma(\theta_k) \in (0, 1)$ are learnable rule weights and the budgets are $B_{\min} = 0.25$, $B_{\max} = 0.50$, $C_1 = 0.15$, $C_2 = 0.25$, $C_3 = 0.15$, $C_4 = 0.10$ and $C_5 = 0.10$.

Those budgets are not arbitrary. They satisfy $B_{\max} + C_2 + C_3 + C_4 = 1$ and $B_{\min} - C_1 - C_5 = 0$, so since every truth value lies in $[0, 1]$ and every weight in $(0, 1)$, the value of Equation (4) lies in $[0, 1]$ for every input and every parameter value and no clipping is ever needed. In the submitted version of this work the identity did not hold, the allocation saturated at one end of its clamp on every pixel of every image, and the rule weights received no gradient, so the rule layer was inert while appearing to work, and a bound enforced by clipping would also make the stability result of Section VII vacuous.

VI. Bounded Change and the Safety Decision

A. Stage five, applying the change

The admitted gain is the raw proposal modulated by the allocation and the strength map, then bounded,

$$\alpha = \text{clamp}(\Delta \odot \mathbf{a} \odot \lambda, -0.20, 0.20). \quad (5)$$

A gate suppresses any change in the dark background,

$$\mathbf{g} = \sigma(20(\mathbf{b} - q_{0.15})), \quad (6)$$

where $q_{0.15}$ is the fifteenth percentile of the backbone output of that image. Using a percentile rather than

a fixed level makes the gate work across scanners with different brightness. A second form of the gate, used where the selection of Section D chose it, is a soft step at the per image Otsu threshold followed by a maximum filter of fixed radius, so that the gate is identically zero beyond that radius from tissue.

The gated gain is applied multiplicatively to give the change $\mathbf{v}$, with the brightening part smoothed before it is added so that it cannot create a new high frequency edge,

$$\mathbf{v} = (\text{smooth}(\text{ReLU}(\boldsymbol{\alpha}) \odot \mathbf{b}) + (\boldsymbol{\alpha} - \text{ReLU}(\boldsymbol{\alpha})) \odot \mathbf{b}) \odot \mathbf{g}, \quad (7)$$

where smooth is a three by three mean filter. The darkening part is left unsmoothed because it has zero mean and cancels itself. In the background, defined by the complement of the gate eroded by two pixels, a masked three by three mean filter is blended in at half strength, which lowers the background standard deviation without letting blur cross into tissue. Finally the edge proposal is added and the result is clipped to the valid range,

$$\mathbf{q} = \text{clamp}(\mathbf{b} + \mathbf{v} + \boldsymbol{\delta}_e \odot \mathbf{g}, 0, 1). \quad (8)$$

B. Stage six, the safety decision

The candidate $\mathbf{q}$ is not returned directly. Three constraints are checked on it.

Aggregate quality must not fall. Define for any image $\mathbf{t}$ the energy

$$E(\mathbf{t}) = 1 - \frac{1}{6} \sum_{i=1}^6 s_i(\mathbf{t}) + 0.1 \sum_{i=1}^6 \max(0, 0.5 - s_i(\mathbf{t})), \quad (9)$$

which is small when the image is good, and whose second term penalises any single property below one half, so a good average cannot hide one broken property. The first constraint requires $E(\mathbf{q}) \leq E(\mathbf{b}) + 0.10$.

No property may be sacrificed for another. The second constraint requires that at least one property improves by more than 0.01 and that no property falls by more than 0.10.

The image must not move far. The third constraint requires $\|\mathbf{q} - \mathbf{b}\|_2 / \|\mathbf{b}\|_2 < 0.2$.

Let $n \in \{0, 1, 2, 3\}$ be the number of constraints that hold. The output is

$$\hat{\mathbf{x}} = (1 - w) \mathbf{b} + w \mathbf{q}, \quad w = \begin{cases} 1, & n \geq 2, \\ 0.05 + 0.10 \frac{n}{3}, & n < 2. \end{cases} \quad (10)$$

Otherwise the output falls back to the backbone while retaining five to eight percent of the proposal, so the decision remains visible rather than silently discarded.

VII. Stability and Guarantees

This section defines the notion of stability that is used, proves that the rule layer has it, and gives the condition

TABLE 2. Here e is the Sobel edge magnitude of the backbone output, σ_k is the local standard deviation over a window of size k , and cv is the local coefficient of variation of the log residual.

P_i	Property	Where the map is large	τ_i
P_1	edge continuity	away from detected edges, where a boundary should continue but does not	0.60
P_2	local contrast	where σ_9 falls below the image mean, and in nearly flat tissue	0.50
P_3	flat smoothness	inside flat regions that still carry residual variation	0.75
P_4	structural coherence	where the edge map is itself rough, a sign of broken structure	0.50
P_5	speckle conformity	where cv departs from the value expected for OCT speckle	0.30
P_6	layer visibility	where the local variance of the edge map is high across depth	0.70

under which the correction cannot reduce contrast. Section I tests both results on real images.

A. What stability means here

This is not stability in the sense of Lyapunov and we claim no fixed point. The quantity that must not behave erratically is the decision of the rule layer, a function of the five rule truth values, which are exactly what shift when the backbone or the scanner changes.

Definition 1 (Stability of the allocation). Let $\mathbf{r} = (r_1, \dots, r_5) \in [0, 1]^5$ be the vector of rule truth values at one pixel and let $a(\mathbf{r})$ be the allocation of Equation (4) at that pixel with the parameters held fixed. The allocation is said to be stable with constant L if

$$|a(\mathbf{r}) - a(\mathbf{r}')| \leq L \|\mathbf{r} - \mathbf{r}'\|_\infty \quad \text{for all } \mathbf{r}, \mathbf{r}' \in [0, 1]^5. \quad (11)$$

In words, a small change in the evidence produces a proportionally small change in the decision, and L is the worst case ratio between the two. Without such a bound the layer could swing from applying no correction to applying all of it because one property score moved slightly, which is precisely the behaviour a clinical user cannot audit.

B. The rule layer is stable, with an exact constant

Theorem 1 (Allocation stability). The allocation of Equation (4) satisfies Definition 1 with

$$L = C_1 w_1 + C_2 w_2 + C_3 w_3 + C_4 w_4 + C_5 w_5, \quad (12)$$

where $w_k = \sigma(\theta_k)$ are the learned rule weights. The constant is attained, so it is the exact Lipschitz constant and not merely an upper bound. Moreover $L < C_1 +$

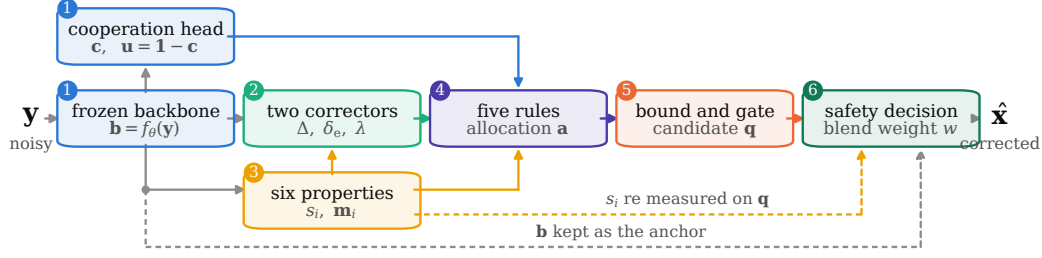


FIGURE 1. The six stages. A frozen backbone and a small head produce the image and the cooperation map. Two networks propose changes. Six properties are measured. Five rules allocate. The change is bounded and gated. Three constraints decide how much is kept.

$C_2 + C_3 + C_4 + C_5 = 0.75$ for every possible setting of the parameters.

The allocation is affine in $\mathbf{r}$ with coefficients $\pm C_k w_k$, so the bound follows from the triangle inequality and is attained when each $r_k - r'_k$ takes the sign of its coefficient. The full proof is given in the supplementary file.

Corollary 1 (Behaviour under a change of backbone). Suppose replacing one backbone by another changes the rule truth values at a pixel by at most ϵ in the maximum norm, with the parameters unchanged. Then the allocation at that pixel changes by at most $L\epsilon$, with L from Equation (12).

This is a statement about the routing rule alone. It says the decision layer degrades gracefully when the evidence shifts, not that image quality transfers, which is the empirical question of Section C.

C. When the correction cannot reduce contrast

Contrast to noise ratio is the measure this method targets most directly. Write $\text{CNR}(\mathbf{t}) = (\mu_{\mathcal{T}}(\mathbf{t}) - \mu_{\mathcal{B}}(\mathbf{t})) / \sigma_{\mathcal{B}}(\mathbf{t})$ for the separation between tissue and background divided by the variability of the background.

Theorem 2 (Contrast margin). Let $\eta_{\mathcal{T}} = \mu_{\mathcal{T}}(\mathbf{q}) - \mu_{\mathcal{T}}(\mathbf{b}) \geq 0$ be the gain in tissue mean and let $\eta_{\mathcal{B}} \geq \mu_{\mathcal{B}}(\mathbf{q}) - \mu_{\mathcal{B}}(\mathbf{b})$ bound the leakage into the background. If the correction does not raise the background variability, that is $\sigma_{\mathcal{B}}(\mathbf{q}) \leq \sigma_{\mathcal{B}}(\mathbf{b})$, then the output of Equation (10) satisfies

$$\text{CNR}(\hat{\mathbf{x}}) \geq \frac{(\mu_{\mathcal{T}}(\mathbf{b}) - \mu_{\mathcal{B}}(\mathbf{b})) + w(\eta_{\mathcal{T}} - \eta_{\mathcal{B}})}{\sigma_{\mathcal{B}}(\mathbf{b})}. \quad (13)$$

In particular, whenever $\eta_{\mathcal{T}} > \eta_{\mathcal{B}}$ the correction cannot reduce the contrast to noise ratio.

The proof is a convex combination argument for the numerator and Minkowski's inequality for the denominator. It is given in full in the supplementary file.

Three parts of the design make the hypotheses of Theorem 2 hold in practice. The gate of Equation (6) keeps $\eta_{\mathcal{B}}$ small by suppressing change in the dark region, the brightness branch of the gain network cannot be negative, which keeps $\eta_{\mathcal{T}}$ non negative, and the background smoothing lowers $\sigma_{\mathcal{B}}(\mathbf{q})$ rather than raising

it. Section I measures all three on every test image and reports how often the hypotheses hold.

Both results are deliberately narrow. Theorem 1 bounds how far the routing decision can move, not whether it is correct, Theorem 2 concerns contrast alone and says nothing about the other five measures, and neither promises that the output is closer to the clean reference, which Section B shows it usually is not. The guarantees concern the policy, not the denoiser underneath it.

VIII. Training

A. What data is used, and where a clean image is needed

Training uses paired data. Each training sample is a noisy scan with a clean reference produced by averaging many repeated acquisitions, and the reference enters two of the five loss terms, the fidelity term and the clinical term, and supervises the additive edge branch. Testing uses no clean image at any point. The six clinical properties of Section A are computed from the noisy input and the backbone output alone, which is what the phrase ground truth free means in this paper. The clean reference scores the results, as in any denoising study, but never reaches the correction path.

B. The five loss terms

Let $\mathbf{x}$ be the clean reference, $\mathbf{b}$ the backbone output and $\hat{\mathbf{x}}$ the wrapper output. The objective of Equation (1) has the following terms.

Fidelity. This term keeps the output near the reference. It combines the squared error with the structural similarity index [22],

$$\mathcal{L}_{\text{fid}} = \|\hat{\mathbf{x}} - \mathbf{x}\|_2^2 + (1 - \text{SSIM}(\hat{\mathbf{x}}, \mathbf{x})). \quad (14)$$

A dead zone is applied to the peak signal to noise ratio part of this term, so no penalty is charged while the drop with respect to the backbone stays below d decibels, which buys clinical quality with a small and bounded amount of fidelity rather than defending fidelity at any price.

Clinical improvement. This term rewards a rise in the clinical measures relative to the backbone, beginning with the contrast to noise ratio and the tissue contrast index, TCI. The logarithm of a ratio makes the term

scale free,

$$\mathcal{L}_{\text{clin}} = -\log \frac{\text{CNR}(\hat{\mathbf{x}})}{\text{CNR}(\mathbf{b})} - \log \frac{\text{TCI}(\hat{\mathbf{x}})}{\text{TCI}(\mathbf{b})}. \quad (15)$$

The same construction is added for the edge preservation index, the equivalent number of looks and the signal to noise ratio, each with its own weight, so a gain in one measure cannot be bought by a collapse in another.

Property calibration. The rule layer compares each property score with a threshold, and if the scores drift away from the thresholds the comparison becomes uninformative, so this term pulls the scores toward the operating point,

$$\mathcal{L}_{\text{pred}} = \sum_{i=1}^6 (s_i(\hat{\mathbf{x}}) - \tau_i)^2. \quad (16)$$

Cooperation. The wrapper should act where the backbone is unsure and stay quiet where it is confident. We measure the agreement between the demand map $\mathbf{u}$ and $\boldsymbol{\pi}$, the summed strength map of the two proposals that the rule layer of Section B reads, with the Pearson correlation ρ , and penalise disagreement,

$$\mathcal{L}_{\text{coop}} = \frac{1}{2} \text{ReLU}(1 - \rho(\mathbf{u}, \boldsymbol{\pi})). \quad (17)$$

The Pearson correlation is differentiable everywhere, invariant to an affine change of either map, and introduces no window size to tune.

Edge supervision. The additive branch is trained to reproduce the gradient that the backbone lost, measured with the Sobel operator ∇ ,

$$\mathcal{L}_{\text{edge}} = \|\nabla \hat{\mathbf{x}} - \nabla \mathbf{x}\|_2^2. \quad (18)$$

C. Optimisation schedule

The backbone is trained first, on the training split with the squared error alone, and is then frozen. The wrapper is then trained in two stages. Only the clinical terms are active at first, so the correctors learn a useful direction, and the fidelity term switches on at the midpoint, so they learn how far they may go. The learning rate groups, the optimiser settings, the full schedule and the procedure as a step by step algorithm are given in the supplementary file.

Throughout both stages the normalisation statistics of f are held fixed. DnCNN is the one backbone with batch normalisation, and left in training mode 45 of its buffers move during fitting, so the frozen reference against which every clinical change of that backbone is measured reads 30.68 dB and a boundary sharpness of 1.031 when the statistics are held fixed against 31.21 dB and 0.948 when they are not. The drift moves the result without touching the wrapper.

D. How the operating point was chosen

The terms of Equation (1) carry fixed relative weights. Two quantities are selected per backbone on the validation split, the fidelity dead zone d of Equation (14) and

the rule of Equation (6) that decides where a correction may act, from the grid listed in the supplementary file, and everything else is held at the value stated in the constants table there for every backbone and every run. For each backbone every setting is trained at three random seeds and scored on the validation images. A setting is admissible only if it clears its bounds at its worst seed rather than on average, and among the admissible settings we take the one with the largest lower confidence bound on its weakest clinical measure. When no setting clears the first pair of bounds we relax them one step at a time and record the step that was needed, so a backbone that required a looser condition is visible rather than hidden. The selected dead zone is 0.6 dB for NAFNet and 0.9 dB for the other three backbones, with the tissue bounded gate for NAFNet and the percentile gate for the others. Every source of randomness is seeded, up to the nondeterminism of the convolution kernels, the spread across seeds is reported where it is available, and the checkpoints chosen on validation are the checkpoints scored on test, with no retraining in between.

E. Initialisation and the choice of every constant

Every constant that carries a design decision is fixed, a structural choice that was not tuned, measured from a property of the data at run time, set once on the validation split while the method was being built, or learned by the objective. Only the two quantities of Section D entered a grid search, and nothing was ever chosen on the test split. The full list, the two deliberate departures from default initialisation, and the structural constants are given in the supplementary file, and Section E reports how much the results move when the constants of the rule layer are changed, the direct test of whether any of them does hidden work.

IX. Tests and Results

A. Setup

Data. Three retinal OCT datasets are used. PKU37 [12] provides 37 cross sections of 640×640 pixels from a spectral domain scanner, with references averaged over 50 frames, split by clean image identity into 27, 5 and 5 for training, validation and testing, giving 1388, 173 and 173 pairs. It is the only training domain. Duke17 [13] provides 16 paired B scans of 970×520 from a clinical scanner at a different institution, and Duke2013 [14] provides 18 volumes of 900×450 with controlled degradation.

Transfer protocol. Every model is trained once on the PKU37 training split and applied to Duke17 and Duke2013 directly, with no parameter or threshold fitted or recalibrated on Duke data and no Duke statistic used to select anything beyond the per image quantity that the gate of Equation (6) computes by the same rule as on PKU37. This no adaptation protocol is the

primary transfer protocol of the paper. The original submission instead used an adapted protocol, in which the PKU37 corrector was trained for fifteen further epochs on the other subjects of the target set in each leave one subject out fold, with the backbone frozen, the wrapper and the cooperation head trainable, and an L2 penalty holding every trainable weight near its PKU37 value. That protocol is rerun for NAFNet and reported in the adapted block of Table 3 and in Section C, and the supplementary file gives its per subject spreads and tests.

Measures. Seven quantities are reported, peak signal to noise ratio against the clean reference for fidelity, and six clinical measures given as percentage change from backbone to corrected output, namely contrast to noise ratio (tissue-background separation), tissue contrast index (dynamic range in tissue), edge preservation index (agreement with the reference edge map), boundary sharpness (gradient across layer boundaries), equivalent number of looks (speckle suppression), and signal to noise ratio (signal against noise floor). Every measure is computed per image then averaged, never pooled across a batch, since pooling inflates the result.

B. In distribution results

Table 3 reports the PKU37 test set and Fig. 2 the visual comparison across the four backbones. Contrast rose on 2064 of the 2076 image and seed combinations. The twelve exceptions all belong to NAFNet, the backbone with the smallest mean contrast gain, fall on ten images whose backbone PSNR lies above the NAFNet mean, and none exceeds 0.4 percent. The gain was larger where the backbone PSNR was lower, with a per image Spearman correlation between minus 0.46 and minus 0.71 across the four backbones. Of the 24 backbone by measure pairs, 21 are positive by more than one seed spread, one is positive but inside it, the tissue contrast index of DnCNN, one is unchanged at 0.00 percent, the edge preservation of KBNNet, and one falls, the boundary sharpness of DnCNN at minus 1.44 percent. The fidelity cost ranges from 0.17 dB on SwinIR to 0.98 dB on DnCNN, and the two backbones that pay least are the two whose six measures all rise.

C. Transfer to other scanners

Table 3 reports the no adaptation protocol on Duke17 and Duke2013. All six clinical measures are positive on both target sets for NAFNet and SwinIR, the fidelity cost stays inside 0.35 dB for those two, and 40 of the 48 backbone by measure by set pairs are positive, the eight exceptions being the tissue contrast index and the boundary sharpness of DnCNN and the edge preservation and the signal to noise ratio of KBNNet. Contrast rises on every backbone and both sets.

The adapted block of Table 3 reports the adapted protocol for NAFNet. Adaptation raises the contrast gain on both scanners, from +7.5 to +10.7 percent on Duke17 and from +7.4 to +15.4 percent on Duke2013, and on Duke2013 it also raises the tissue contrast index from +1.4 to +5.1 percent. It pays for this on both scanners in fidelity, a further 0.06 and 0.10 dB, and in signal to noise ratio, which falls from +5.4 to +2.9 percent on Duke17 and from +9.3 to −5.3 percent on Duke2013, a decrease on 15 of 18 subjects, while the remaining measures move in opposite directions on the two sets. Adaptation therefore reliably improves only the measure the frozen model already improves, so the frozen model is the one we recommend and use everywhere else.

D. Contribution of each clinical property

One property at a time is switched off at test time, by setting its score to the satisfied value and its failure map to zero, and the whole test set is re evaluated. Only P_5 , P_1 and P_2 move any measure by more than a quarter point. P_5 costs the most at 3.19 points of the speckle gain, P_1 1.72 and P_2 0.74, while P_3 , P_4 and P_6 change nothing by more than 0.23 points, so half the set is close to inert here. We keep all six, because they are the vocabulary of the rule layer and of the explanation shown to the reader, and one that rarely fires here can fire on another scanner.

E. Sensitivity to the rule constants

The rule layer contains a base level, five rule weights and five budgets. The budgets are tied by the identity of Section B, so varying one alone would reintroduce the defect that identity prevents. We therefore vary the base level and the two rule weights that carry the allocation, and we replace the Lukasiewicz conjunction by the product and by the minimum, the two other standard choices, to test whether the specific logic matters. No variation turns any of the six clinical measures negative. Across the fourteen variations the contrast gain lies between 3.5 and 4.5 percent, the tissue contrast index between 2.3 and 2.9 percent, the speckle measure between 22.6 and 23.5 percent and the fidelity cost between 0.16 and 0.23 dB. The base level is the only constant whose effect is visible, since lowering it raises contrast and costs fidelity together, while the two rule weights move the result by under a fifth of a point. The product and the minimum change the contrast gain from 3.7 to 3.5 percent and leave the other five measures within one part in fifty, so the result rests on the shape of the rule layer and not on the particular t norm, which limits what the specific logic can be credited with as much as it reassures. No setting of either gate rule of Section D recovers boundary sharpness on DnCNN.

TABLE 3. PKU37 test set of 173 images, then Duke17 and Duke2013 with no adaptation, then the adapted protocol of the original submission for NAFNet described in Section A. The first numeric column is the backbone peak signal to noise ratio in decibels and the second its change after correction with the spread across three seeds. The six clinical measures are percentage change from backbone output to corrected output. Bold marks a clinical cell whose mean exceeds its seed spread. The adapted rows come from one seed and one fold per subject, so they carry no spread and no bold. The two lowest blocks use the selected NAFNet checkpoint on the same 173 test images with one component removed or an alternative operator substituted, so those rows differ only in that. Full spreads are given in the supplementary file.

Backbone	Backbone PSNR (dB)	Δ PSNR (dB)	Δ CNR	Δ TCI	Δ EPI	Δ BS	Δ ENL	Δ SNR
PKU37 test set, 173 images								
NAFNet	30.75	-0.18 $\pm$ 0.02	+3.49	+1.91	+1.20	+3.89	+23.74	+11.10
DnCNN	30.68	-0.98 $\pm$ 0.21	+11.30	+0.99	+4.14	-1.44	+57.14	+22.95
SwinIR	29.87	-0.17 $\pm$ 0.06	+7.33	+11.45	+2.68	+1.82	+28.32	+11.54
KBNet	32.04	-0.90 $\pm$ 0.08	+6.93	+5.92	+0.00	+2.58	+25.30	+7.95
Duke17, no adaptation								
NAFNet	27.03	-0.17 $\pm$ 0.01	+7.51	+4.00	+1.37	+4.15	+21.35	+5.43
DnCNN	26.89	-0.29 $\pm$ 0.05	+12.19	-1.47	+3.49	-2.24	+39.42	+13.55
SwinIR	27.06	-0.19 $\pm$ 0.05	+8.68	+13.26	+2.69	+1.90	+29.46	+11.60
KBNet	27.59	-0.51 $\pm$ 0.05	+5.34	+7.19	-0.50	+1.61	+7.49	-1.81
Duke2013, no adaptation								
NAFNet	27.33	-0.35 $\pm$ 0.01	+7.37	+1.41	+0.85	+0.51	+28.80	+9.29
DnCNN	27.21	-0.42 $\pm$ 0.05	+12.73	-2.21	+2.99	-5.85	+49.91	+17.59
SwinIR	26.89	-0.19 $\pm$ 0.06	+9.26	+11.86	+2.55	+2.30	+30.90	+13.31
KBNet	28.12	-0.59 $\pm$ 0.04	+7.16	+4.83	-0.35	+3.06	+8.70	-0.90
NAFNet on PKU37, one component removed								
Full method		-0.17	+3.71	+2.42	+1.21	+4.16	+23.34	+10.74
No rule layer		-0.38	+0.83	+0.38	+0.99	+4.27	+25.45	+10.44
No edge branch		-0.37	-2.71	-6.54	+0.55	+0.65	+20.28	+7.30
No cooperation map		-0.15	+2.51	+1.15	+1.10	+3.87	+23.84	+10.61
No background smoothing		-0.18	+3.03	+6.88	+0.92	+3.99	+5.75	+2.85
NAFNet on PKU37, alternatives of similar complexity on the same backbone output								
Uniform allocation, trained		-0.42	+4.11	+11.58	+0.71	+3.83	+16.18	+5.80
Unsharp masking		-0.67	+0.63	+39.45	-3.59	+25.13	-35.31	-20.57
Adaptive equalisation		-4.43	+6.48	+53.03	-3.45	+24.62	-38.60	-25.13
Adapted on the target set, leave one subject out, NAFNet only								
NAFNet, Duke17	27.03	-0.23	+10.68	+2.69	+1.49	+4.13	+24.29	+2.95
NAFNet, Duke2013	27.33	-0.45	+15.43	+5.15	+0.90	+2.96	+9.23	-5.29

F. Contribution of each component

The component block of Table 3 removes one component at a time. Replacing the rule layer by a uniform allocation cuts the contrast gain from 3.7 to 0.8 percent and the tissue contrast index from 2.4 to 0.4 percent while doubling the fidelity cost, the clearest evidence that the allocation and not the capacity of the correctors produces the clinical gain. Removing the edge branch turns both measures negative, and removing the cooperation map costs about a third of the contrast gain. Removing background smoothing leaves contrast, edge preservation and boundary sharpness almost intact while the speckle measure collapses from 23.3 to 5.8 percent and the signal to noise ratio from 10.7 to 2.9 percent. The four components are therefore not interchangeable, and each carries a different measure.

G. Comparison at matched complexity

Three alternatives act on the same frozen backbone output. The first is the same architecture, budget, objective and ten epochs, with the rule layer replaced by a uniform allocation for the whole of training rather than only at test time, keeping the properties and the safety layer. It was fitted before the gate search, so it carries the percentile gate rather than the tissue bounded gate NAFNet was finally selected with, which we state rather than claim a match we cannot verify. The other two are classical operators, unsharp masking and contrast limited adaptive histogram equalisation, tuned on the validation split. Uniform allocation reaches a larger contrast gain and a much larger tissue contrast index than the full method, for 0.42 dB against 0.17 dB, and is worse on four of the six measures, so the two

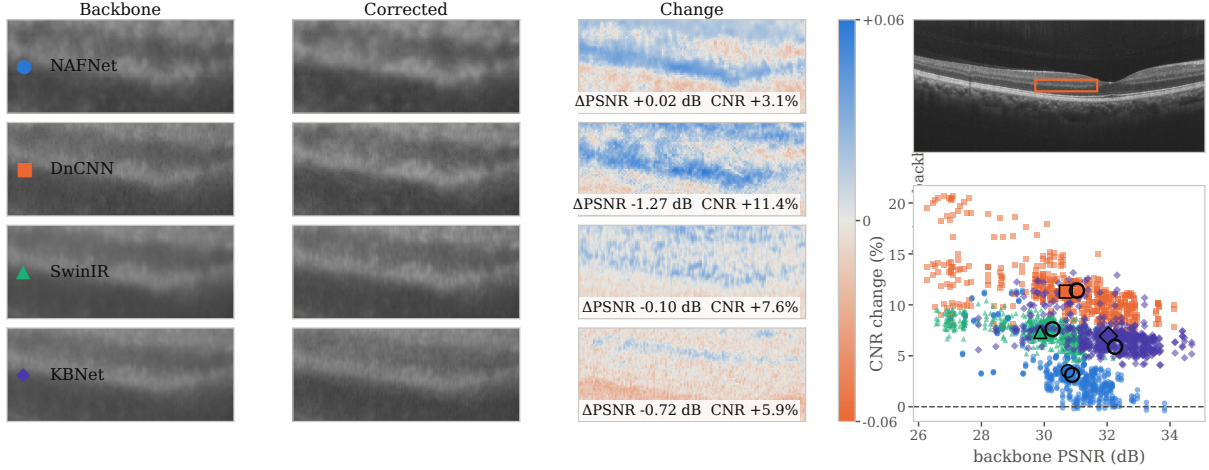


FIGURE 2. Visual comparison on the PKU37 test image of median backbone fidelity, one row per backbone, in the 56 by 136 pixel band marked on the thumbnail where the NAFNet correction was largest. Cells show the backbone output and the corrected output on the display range 0.05 to 0.85, and their signed difference on a fixed scale of plus or minus 0.06. The mean absolute change is about one percent of the intensity range, so the two outputs are near identical by design and the difference column shows where and how much each backbone is corrected, most for SwinIR and least for KBNet. All values are from the seed 0 checkpoints of Table 3. The scatter places this image, ringed, among all 173 test images at three seeds for four backbones, with the twelve NAFNet points below zero marked. Full resolution pairs for every backbone are in the supplementary file.

are not ordered. What the full method buys with its fidelity is the per pixel account of why each change was made. The classical operators give the largest contrast numbers and fail elsewhere, lowering edge preservation by about 3.5 percent and the speckle measure by about 36 percent while raising boundary sharpness by about 25 percent by sharpening noise with structure, with adaptive equalisation also costing 4.4 dB. Every alternative that produces larger numbers changes the image several times as much, unbounded and unexplained.

H. Does the cooperation map predict error

Called a confidence map in the original submission, this component was tested against the absolute error of the backbone output over 173 images, and it does not predict that error. The rank correlation is -0.258, so the map is ordered the wrong way rather than merely uninformative, and removing the pixels it ranks as least reliable raises the remaining error to 1.88 times the starting value where an oracle ranking reaches 0.07, a sparsification error of 0.776. The map is nevertheless load bearing, since replacing it by a constant lowers the contrast gain from +3.71 to +2.51 percent and worsens five of the six clinical measures. The head is shown only where a correction helps and never the error of the backbone, so nothing in the objective could have made it a confidence, and we call it a cooperation map rather than a calibrated one.

I. Do the theoretical conditions hold on real data

We perturb the rule truth values and compare the ratio of Equation (11) with the constant of Equation (12), and we measure the tissue gain, the background leakage and the ratio of background standard deviations on every

image against the hypotheses of Theorem 2. The stability bound holds with room to spare, a constant of 0.5489 from Equation (12) against a largest measured ratio of 0.4679 over 173 images, so the proof is conservative rather than tight. The allocation map spans 0.556 to 0.793 with mean 0.726 and standard deviation 0.050, and 0.00 percent of pixels sit at either end of its clamp, direct evidence that the rule layer does work after the repair of Section B.

The contrast hypotheses hold on a majority of images and not on all. The margin condition holds on 139 of 173 images and the standard deviation ratio condition on 94, a bare majority, with mean ratio 0.976, below one, which is why the result holds in the mean. Where a hypothesis fails the theorem makes no prediction, and the measured contrast change there is not systematically different in sign, so the conditions are sufficient rather than necessary. Both leakage terms of Theorem 2 sit far below the mean absolute change over the whole image, 0.0067, at -0.0007 in tissue and -0.0017 in background. Their means are near zero with a long negative tail, so the per image sign carries what the theorem needs. Tissue is positive on 137 of 173 images and background negative on 107, the direction the decomposition assumes.

J. Safety, and whether the correction invents structure

A correction that raises contrast could sharpen a boundary that is not there or erase a faint one that is, and the PKU37 clean reference lets us measure both. Two measures move against the correction. An edge appears where the reference has none on 0.48 percent of pixels after correction against 0.45 percent for the backbone output, a rise on 127 of 173 images with $p = 1.9 \times 10^{-12}$ under a two sided Wilcoxon signed rank test on the

paired per image values, and the retention of structures whose reference contrast falls in the lowest decile falls from 40.5 to 37.6 percent, on 168 of 173 images with $p = 6.2 \times 10^{-30}$. Both changes are small, 0.03 points of invented edge rate and 2.9 points of retention, and the clamp of Section B bounds how far any pixel may move but not how many pixels cross an edge threshold, so neither is zero by construction and Section X lists both as limitations. What separates the correction from a sharpening filter is therefore Section G, where the two classical sharpeners lose edge preservation and the speckle measure while the wrapper gains on both. The remaining measures move with the correction. The two layer boundaries a clinician reads are not displaced, with a mean absolute shift of 4.73 to 4.38 pixels for the inner limiting membrane, an improvement, and 9.93 to 10.03 for the retinal pigment epithelium, unchanged within the spread across images. Dark fluid like regions occupy 1.6 percent of the image area with a mean absolute change of 0.0022 inside them against 0.0067 over the whole image, and the largest change on any single image anywhere in the test set is 0.0178, so the correction is quieter where a hypodense region carries the finding. Boundary sharpness is not worsened on 104 of 173 images, reported as a count because the per image worst case is what matters for trusting a scan. Of the three constraints of Section B, energy and bounded change hold on all 173 images and only the no sacrifice constraint ever fails, on 38 of them. Acceptance needs two of three and no image falls below two, so the blend of Equation (10) returns the candidate in full everywhere, so the safety layer never had to act here.

K. Cost of moving to a new backbone

A new backbone means refitting the whole wrapper, not only the head that reads its features. The trainable count is 168 630 shared parameters plus a head whose width follows the feature tensor it reads, 175 893 in all for NAFNet and 254 538 for SwinIR, against about seven million in the backbone, an increase of about 2.4 percent. Every backbone is fitted on the same 1388 PKU37 pairs for the same ten epochs, so only the wall clock time and the spread across seeds vary. Fitting one seed of the selected setting on a single RTX 5070 Ti takes 2.0 minutes for NAFNet, 2.5 for KBNNet, 3.9 for DnCNN and 8.3 for SwinIR, an ordering that follows the cost of one backbone forward pass and not anything in the wrapper. The spread of the fidelity cost across the three seeds is 0.02 dB for NAFNet, 0.06 for SwinIR, 0.08 for KBNNet and 0.21 for DnCNN. The backbone with the widest spread is also the only one with a clinical measure that falls, so a new user should read the seed spread as the first warning sign rather than the mean. Adoption therefore costs under ten minutes of fitting per seed, no

change to the backbone, and one sweep of the fidelity dead zone.

X. Discussion

A. Why fidelity falls, and when that is the right trade

Peak signal to noise ratio falls when the wrapper is applied, by 0.17 to 0.98 dB depending on the backbone. A denoiser trained on squared error sits at a local optimum of that error, so any change that raises contrast or boundary definition moves away from it, and the question is when the trade is worth making. A change of a few tenths of a decibel is smaller than the variation between two acquisitions of the same eye, while the quantities that improve are what a grader uses to place a boundary or judge a fluid pocket. The size of the gain tracks the room a backbone leaves on each measure rather than its overall fidelity. On boundary sharpness NAFNet starts lowest at 0.851 of the reference and gains most, and DnCNN starts at 1.031, above the reference, and is the only backbone that falls. On edge preservation KBNNet starts highest at 0.835 and is the only backbone left unchanged. For a deployment that must not lose fidelity at all, tightening the third constraint of Section B moves the output smoothly toward the backbone output. SwinIR is the instructive case, since windowed attention smooths hard inside a window, so it leaves the most room on the contrast measures and yet pays the least fidelity of the four at 0.17 dB.

B. Limitations

The correction is bounded by construction, so a backbone that has destroyed a structure cannot be made to recover it. Two safety measures move against it on the test set. The invented edge rate rises from 0.45 to 0.48 percent and the retention of the weakest decile of structures falls from 40.5 to 37.6 percent, on 127 and 168 of 173 images. Both are small and bounded in size, neither is zero, and the validation criterion that selected the operating point scores six clinical measures and fidelity and does not see either of them. We measured what that blindness costs, on the validation split so that the test split is used once. The setting the criterion ranked second for NAFNet raises weak structure retention from 40.7 to 43.9 percent where the selected setting lowers it to 37.7, and adds less to the invented edge rate. The criterion is therefore blind in a way that matters, and a user who values faint detail above contrast should score these two measures during selection. We did not reselect on this evidence, because the criterion was fixed before the test split was scored and changing it afterwards would spend that split twice. The formal results are narrow in the way stated at the end of Section VII. Moving to a new backbone means refitting the whole wrapper, at the cost Section K reports. The clinical measures used here are established proxies, not outcomes, and no clinician

read the images, so the safety analysis of Section J is quantitative evidence rather than clinical validation. Finally, the external datasets are small, sixteen and eighteen images, so the transfer results are evidence of behaviour across scanners rather than a precise estimate of the gain on any one device.

XI. Conclusion

We presented a post correction layer for OCT denoising that is attached to a frozen denoiser rather than replacing it. It states six clinical properties in closed form, turns them into a per pixel decision through five named rules, and limits how far the resulting change may move the image, so a change can be traced back to the property that motivated it and the constraint that allowed it to stand. Across four backbone families the same rules with the same fixed weights raise the clinical measures at a fidelity cost between 0.17 and 0.98 dB, and no measure falls on three of the four. The exception is boundary sharpness on DnCNN, which falls by 1.4 percent from a value that already exceeded the clean reference, showing where the method has no room to act. The measures are proxies, and the next step is a reader study asking whether the changes the method makes are the changes a grader would want.

REFERENCES

- [1] D. Huang et al., "Optical coherence tomography," *Science*, vol. 254, no. 5035, pp. 1178–1181, 1991.
- [2] J. W. Goodman, *Speckle Phenomena in Optics: Theory and Applications*. Roberts & Company, 2007.
- [3] S. K. Devalla et al., "DRUNET: A dilated-residual U-Net deep learning network to segment optic nerve head tissues in optical coherence tomography images," *Biomed. Opt. Express*, vol. 9, no. 7, pp. 3244–3265, 2018.
- [4] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, "Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising," *IEEE Trans. Image Process.*, vol. 26, no. 7, pp. 3142–3155, 2017.
- [5] L. Chen, X. Chu, X. Zhang, and J. Sun, "Simple baselines for image restoration," in *Proc. ECCV*, 2022, pp. 17–33.
- [6] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van Gool, and R. Timofte, "SwinIR: Image restoration using Swin Transformer," in *Proc. ICCV Workshops*, 2021, pp. 1833–1844.
- [7] A. Krull, T.-O. Buchholz, and F. Jug, "Noise2Void—learning denoising from single noisy images," in *Proc. CVPR*, 2019, pp. 2129–2137.
- [8] E. van Krieken, E. Acar, and F. van Harmelen, "Analyzing differentiable fuzzy logic operators," *Artif. Intell.*, vol. 302, art. 103602, 2022.
- [9] Y. Zhang et al., "KBNet: Kernel basis network for image restoration," *Tech. Rep. arXiv:2303.02881*, 2023. [Online]. Available: <https://arxiv.org/abs/2303.02881>
- [10] A. Buades, B. Coll, and J.-M. Morel, "A non-local algorithm for image denoising," in *Proc. CVPR*, vol. 2, 2005, pp. 60–65.
- [11] M. A. Mayer, A. Borsdorf, M. Wagner, J. Hornegger, C. Y. Mardin, and R. P. Tornow, "Wavelet denoising of multiframe optical coherence tomography data," *Biomed. Opt. Express*, vol. 3, no. 3, pp. 572–589, 2012.
- [12] M. Geng et al., "Triplet cross-fusion learning for unpaired image denoising in optical coherence tomography," *IEEE Trans. Med. Imag.*, vol. 41, no. 11, pp. 3357–3372, 2022.
- [13] L. Fang, S. Li, Q. Nie, J. A. Izatt, C. A. Toth, and S. Farsiu, "Sparsity based denoising of spectral domain optical coherence tomography images," *Biomed. Opt. Express*, vol. 3, no. 5, pp. 927–942, 2012.
- [14] L. Fang et al., "Fast acquisition and reconstruction of optical coherence tomography images via sparse representation," *IEEE Trans. Med. Imag.*, vol. 32, no. 11, pp. 2034–2049, 2013.
- [15] J. Wang et al., "Deep learning for quality assessment of retinal OCT images," *Biomed. Opt. Express*, vol. 10, no. 12, pp. 6057–6072, 2019.
- [16] Y. Gal and Z. Ghahramani, "Dropout as a Bayesian approximation: Representing model uncertainty in deep learning," in *Proc. ICML*, 2016, pp. 1050–1059.
- [17] A. Kendall, Y. Gal, and R. Cipolla, "Multi-task learning using uncertainty to weigh losses for scene geometry and semantics," in *Proc. CVPR*, 2018, pp. 7482–7491.
- [18] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Proc. NeurIPS*, 2017, pp. 6402–6413.
- [19] A. d'Avila Garcez and L. C. Lamb, "Neurosymbolic AI: The 3rd wave," *Artif. Intell. Rev.*, vol. 56, pp. 12387–12406, 2023.
- [20] D. M. Stein et al., "A new quality assessment parameter for optical coherence tomography," *Br. J. Ophthalmol.*, vol. 90, no. 2, pp. 186–190, 2006.
- [21] P. Tewarie et al., "The OSCAR-IB consensus criteria for retinal OCT quality assessment," *PLoS ONE*, vol. 7, no. 4, art. e34823, 2012.
- [22] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, 2004.
- [23] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. ICLR*, 2019.
- [24] R.-E. Precup, C.-A. Bojan-Dragos, E.-L. Hedrea, R.-C. Roman, and E. M. Petriu, "Evolving fuzzy models of shape memory alloy wire actuators," *Rom. J. Inf. Sci. Technol.*, vol. 24, no. 4, pp. 353–365, 2021.
- [25] Y. Fan, H. Wang, H. Gemmeke, T. Hopp, and J. Hesser, "Model-data-driven image reconstruction with neural networks for ultrasound computed tomography breast imaging," *Neurocomputing*, vol. 467, pp. 10–21, 2022.
- [26] M. Casquilho, R. Galhano, and J. Buescu, "Effect of random variation of the initial values of ordinary differential equations and its web-based application to chemical kinetics," *Rom. J. Inf. Sci. Technol.*, vol. 28, no. 3, pp. 286–298, 2025.
- [27] Z. Han, B. Wei, X. Xi, B. Chen, Y. Yin, and S. Li, "Unifying neural learning and symbolic reasoning for spinal medical report generation," *Med. Image Anal.*, vol. 67, art. 101872, 2021.
- [28] A. Nguembang Fadja, M. Fraccaroli, A. Bizzarri, G. Mazzuchelli, and E. Lamma, "Neural-symbolic ensemble learning for early-stage prediction of critical state of Covid-19 patients," *Med. Biol. Eng. Comput.*, 2022.
- [29] Q. Chen, L. Zeng, and C. Lin, "A deep network embedded with rough fuzzy discretization for OCT fundus image segmentation," *Sci. Rep.*, 2023.
- [30] T.-W. Kim and K.-C. Kwak, "Pixel-level fuzzy rule attention maps for interpretable MRI classification," *Symmetry*, vol. 17, no. 12, art. 2187, 2025.
- [31] A. Özkan, Y.-H. Li, and T. Sikora, "Steered-mixture-of-experts regression for image denoising with multi-model-inference," in *Proc. 31st Eur. Signal Process. Conf. (EU-SIPCO)*, 2023, pp. 546–550.
- [32] A. Özkan, Y.-H. Li, and T. Sikora, "Denoising OCT images using steered mixture of experts with multi-model inference," in *Proc. SPIE*, vol. 12830, art. 1283009, 2024.
- [33] E. Zamfir et al., "Complexity experts are task-discriminative learners for any image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025, pp. 12753–12763.
- [34] M. H. Wang and S. Qin, "Explainable neuro-symbolic artificial intelligence for automated interpretation of corneal topography and early keratoconus detection," *Front. Artif. Intell.*, vol. 9, art. 1713747, 2026.